

SHRUTI C. NANIVADEKAR

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EDUCATION

University of Pittsburgh School of Medicine, Pittsburgh, PA

expected 2027

Medical Scientist Training Program Fellow

Clerkship grades: Honors - IM, Surgery, Psych; High Pass - Neurology, ObGyn, Pediatrics

Carnegie Mellon University, Pittsburgh, PA

June 2025

Ph.D. in Systems Neuroscience

Thesis: Characterizing network-specific pathophysiology to guide targeted therapeutic interventions for motor recovery in parkinsonian motor syndromes

University of California, Los Angeles, CA

June 2016

Bachelor of Science in Neuroscience, *summa cum laude*

Overall GPA: 3.94/4.00

RESEARCH EXPERIENCE

Gittis Lab for Neural Basis of Motor Control, CMU

June 2020-June 2025

Graduate Student Researcher, Advisor: Aryn Gittis

Pittsburgh, PA

- Developed an in-vivo model of electrical stimulation in mice with bilateral dopamine depletion
- Utilized optogenetically-inspired cell-type modulation to identify patterns and targets of electrical deep brain stimulation (DBS) to enhance motor recovery in Parkinson's disease (PD)
- Investigated the neural activity in the thalamus of two models of PD pathology that show either tremor or bradykinesia

Turner lab for Basal-Ganglia Disorders, University of Pittsburgh

June 2018-August 2018

Summer Rotation Student

Pittsburgh, PA

- Collected single-unit neural recording data from the subthalamic nucleus of a parkinsonian non-human primate
- Applied spike-field analysis to analyze the movement-related coherence between GPi and thalamic LFP activity in simultaneous neural recordings

National Institutes of Neurological Disorders and Stroke, NIH

September 2016-May 2018

Postbaccalaureate Intramural Research Training Award Fellow

Bethesda, MD

- Analyzed diffusion tensor imaging (DTI) in patients with PD to understand the white matter correlates of the cognitive side effects of DBS
- Developed tractography-based predictor for side effect profile of various stimulation parameters

Staba lab for Epilepsy, UCLA

July 2013-June 2016

Research Assistant

Los Angeles, CA

- Developed an automated seizure detection algorithm in MATLAB for EEG data from post-traumatic rats
- Validated automated detection of seizure spread activity across multiple depth electrodes

Blanke Lab of Cognitive Neuroscience, EPFL

June 2015-August 2015

Summer Research Intern

Lausanne, Switzerland

- Investigated multisensory metacognition by comparing judgement in visual and auditory behavioral tasks
- Collected EEG, ECG, and behavioral data, analyzed in MATLAB and R to estimate metacognitive performance

O'Neill lab for Child and Adolescent Psychiatry, UCLA

March 2013-December 2013

Research Assistant

Los Angeles, CA

- Collected EEG data from subjects diagnosed with obsessive compulsive disorder, processed using Independent Component analysis and Dipole Source Localization

PEER-REVIEWED PUBLICATIONS

Nanivadekar, S., Salas-Allende, I., Nuber, S., Turner, R.S., Gittis, A.H. (2026). *Aberrant Cerebellar-Recipient Thalamic Activity in Two Mouse Models with Prominent Tremor or Bradykinesia*. **The Journal of Neuroscience**, 46(19), e1630252026.

Spix, T. A., **Nanivadekar, S.**, Toong, N., Kaplow, I. M., Isett, B. R., Goksen, Y., Pfenning, A. R., Gittis, A. H. (2021). *Population-specific neuromodulation prolongs therapeutic benefits of deep brain stimulation*. **Science**, 374(6564), 201-206.

Kariv, S., Choi, J. W., Mirpour, K., Gordon, A. M., Alijanpourtaghsara, A., Benam, M., Abdalla, R., Chilukuri, S., Gu, J.W., Bokil, H., **Nanivadekar, S.**, Gittis, A.H., Pouratian, N. (2024). *Pilot study of acute behavioral effects of pallidal burst stimulation in Parkinson's disease*. **Movement Disorders**, 39(10), 1873-1877.

Yang, Q., **Nanivadekar, S.**, Taylor, P. A., Dou, Z., Lungu, C. I., Horovitz, S. G. (2021). *Executive function network's white matter alterations relate to Parkinson's disease motor phenotype*. **Neuroscience letters**, 741, 135486.

CONFERENCE PRESENTATIONS

S. Nanivadekar, A. Gittis. 2025. *Circuit-inspired strategies to improve treatments for Parkinson's disease*. NANS Neural Interfaces, Washington DC

S. Nanivadekar, A. Gittis. 2023. *Characterization of behavioral impairments and effects of DBS in the 3K tremor mouse model*. Society for Neuroscience, Washington DC

S. Nanivadekar, T. Spix, A. Gittis. 2023. *Cell-type Specific Neuromodulation Using Burst DBS Produces Long-lasting Behavioral in a Parkinsonian Mouse Model*. Lake Conference Sensation and Action, Thun, Switzerland

S. Nanivadekar, T. Spix, A. Gittis. 2021. *Burst DBS induces persistent motor rescue in parkinsonian mice*. Society for Neuroscience, Virtual

S. Nanivadekar, D. Kase, R. Turner. 2018. *Mechanisms of neural communication in the basal ganglia-thalamocortical circuit in a parkinsonian monkey*. Medical Scientist Training Program Retreat, Pittsburgh

LEADERSHIP EXPERIENCE

Student Committee Chair, Medical Scientist Training Program June 2022-June 2023

- Served as a liaison between MSTP students and administration to enhance student experiences in the program
- Coordinated with all student committees to facilitate execution of most MSTP activities and events

Retreat Chair, Mentorship Committee, Medical Scientist Training Program June 2020-June 2021

- Organized the annual retreat, involving tasks like finding keynote speakers, hosting a scientific symposium with student talks and poster sessions, and planning activities that foster mentorship and professional development
- Organized social events within the program to facilitate integration across MSTP classes

GRANTS AND AWARDS

NIH National Institute of Neurological Disorders and Stroke F31 NRSA, CMU	2022-2025
BS Neuroscience, Highest Departmental Honors, UCLA	2016
Vice Provost Prize for Research Article in Undergraduate Science Journal, UCLA	2016
Deans Honors List, UCLA	2012-2016

TECHNICAL STRENGTHS

Computer Languages and Modeling tools
Application Software
Languages

MATLAB, Java, R, STATA, C++, Python
Affinity Designer, Arduino IDE, BlueJ IDE, Microsoft Office
Fluent in English, Hindi and Marathi